
Final Project Report: Advanced Chooser Option Pricing Model with World Data and Machine Learning

Integrated final report synthesizing baseline chooser pricing, sensitivity analysis, machine-learning volatility prediction, ML-enhanced BSM pricing, supplementary diagnostics, and the Streamlit prototype.

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1. Introduction

Chooser options give the holder the right to decide at a future choice date whether the contract should become a call or a put. This flexibility makes the instrument useful in structured products, risk management, and quantitative trading settings where the direction of the underlying asset is uncertain but convex exposure remains valuable.

The project objective was to reproduce and validate a Black-Scholes-Merton based chooser option pricing framework, enrich it with real-world market data and engineered features, use machine learning to improve forward-looking volatility prediction, test whether an ML-enhanced sigma improves pricing performance, and deliver a usable prototype tool. The final workflow therefore combines classical pricing theory, empirical robustness checks, machine-learning diagnostics, and a practical dashboard layer.

2. Pricing Framework

The baseline pricing framework follows the BSM logic for a chooser option with a fixed strike, maturity, chooser decision time, risk-free rate, dividend yield, and volatility input. The implementation prices the call and put components consistently under the same market assumptions and uses the chooser structure to capture the value of the later exercise-style decision.

The central empirical limitation of the baseline framework is the volatility input. The baseline route uses a historical-volatility style sigma derived from observed JPM price behavior. A core goal of the project was therefore to replace or enhance this input with a machine-learning forecast of future realized volatility while keeping the BSM pricing structure fixed. This isolates the question: does a better forward-looking sigma translate into better chooser pricing?

Figure 1. Baseline and ML-Enhanced Chooser Price Trend



Figure 1 compares historical chooser prices under the baseline BSM sigma and the final ML-enhanced sigma.

3. Data and Feature Construction

The project uses processed JPM equity data, option benchmark data, Treasury-rate data, VIX data, and engineered time-series features. The processed CSV outputs serve as the reproducible interface between data preparation, pricing, model training, sensitivity analysis, and report generation.

The engineered feature set includes realized volatility, rolling stock volatility, volatility-of-volatility, Treasury-rate changes and momentum, VIX returns, Treasury levels, and rolling JPM-VIX correlation features. These variables were designed to capture both local stock dynamics and broader risk-regime information.

For the ML workflow, the target variable is future realized volatility over a 20-day horizon. The data are split into train, validation, and test partitions, allowing model selection to be separated from final out-of-sample evaluation.

Table 1. Processed Dataset Coverage

Source: data/processed/ml_volatility_train.csv; data/processed/ml_volatility_val.csv; data/processed/ml_volatility_test.csv; data/processed/final_ml_enhanced_bsm_prices.csv
Compact row counts and date ranges; raw data are not dumped.

Dataset	Source	Rows	Start	End
Train	...	1176	2018-04-26	2022-12-23
Validation	...	252	2022-12-27	2023-12-27
Test	...	253	2023-12-28	2024-12-30
Pricing history	...	1681	2018-04-26	2024-12-30

4. Baseline Pricing and Robustness Analysis

The baseline BSM chooser implementation provides the reference pricing path. Robustness analysis then asks how stable this pricing logic is under market shocks and market regimes. Week 7 sensitivity experiments show that chooser prices are strongly convex to volatility changes, while the tested rate-hike shock has a smaller and negative effect.

The extreme-scenario tables summarize base, volatility-spike, rate-hike, and combined-shock behavior. The regime tables compare normal and stress-period responses, making clear that stress periods raise the level and shock sensitivity of chooser prices.

Table 2. Mean Chooser Price by Extreme Scenario

Source: outputs/week7_extreme_scenario_summary.csv

Scenario	MeanPrice
Baseline Base	38.6525
Baseline Vol Spike	46.6612
Baseline Rate Hike	37.8280
Baseline Combined Shock	45.8588
ML Base	38.6215
ML Vol Spike	46.6151
ML Rate Hike	37.7943
ML Combined Shock	45.8098

Table 3. Mean Change Relative to Base Scenario

Source: outputs/week7_extreme_scenario_change_summary.csv

Scenario	MeanChange_vs_Base
Baseline Vol Spike	8.0087
Baseline Rate Hike	-0.8246
Baseline Combined Shock	7.2062
ML Vol Spike	7.9936
ML Rate Hike	-0.8273
ML Combined Shock	7.1883

Table 4. Regime-Level Mean Chooser Prices

Source: outputs/week7_regime_summary.csv

Regime	Baseline_Base	Baseline_VolSpi	Baseline_RateHi	Baseline_Combi	ML_Base	ML_VolSpike	ML_RateHike	ML_CombinedShock
Normal	35.5026	42.7217	34.9344	42.1367	35.5124	42.7549	34.9424	42.1679
Stress_2020	56.4316	68.8970	54.1603	66.8674	56.1705	68.4035	53.8913	66.3661

Table 5. Regime-Level Mean Changes

Source: outputs/week7_regime_change_summary.csv

Regime	Baseline_VolSpike	Baseline_RateHik	Baseline_Combine	ML_VolSpike_Chai	ML_RateHike_Cha	ML_CombinedShock_Change
Normal	7.2191	-0.5682	6.6341	7.2425	-0.5700	6.6555
Stress_2020	12.4654	-2.2713	10.4358	12.2330	-2.2793	10.1956

Figure 2. Mean Price under Extreme Scenarios

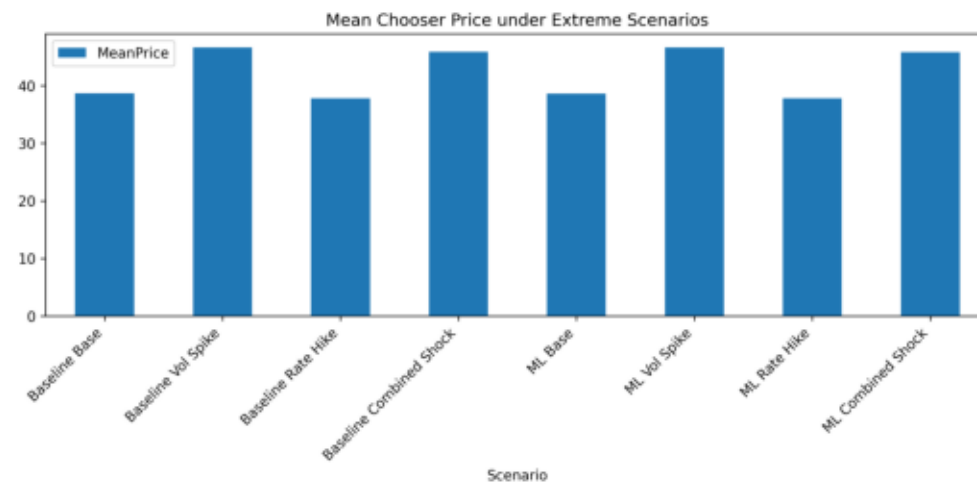


Figure 2 shows the level effect of volatility and rate shocks across baseline and ML-enhanced BSM prices.

Figure 3. Mean Change versus Base

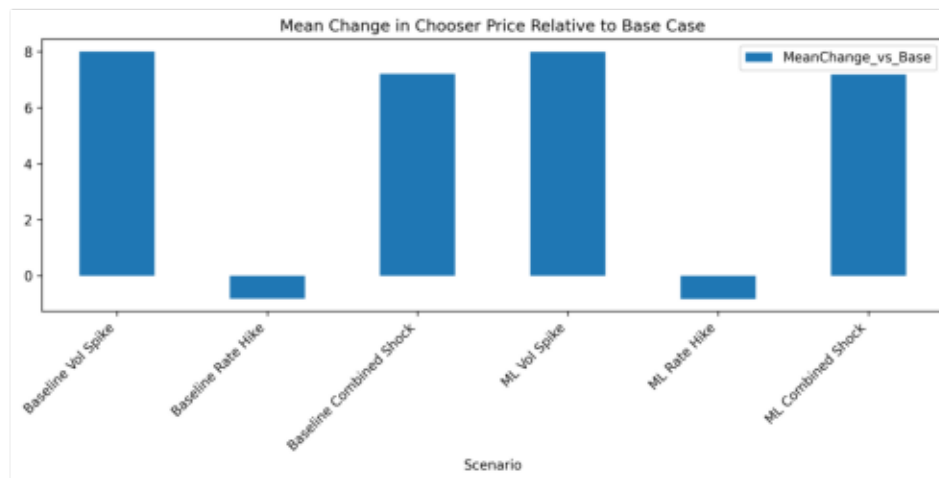


Figure 3 shows that volatility shocks dominate repricing relative to the base case.

Figure 4. Volatility Shock by Regime

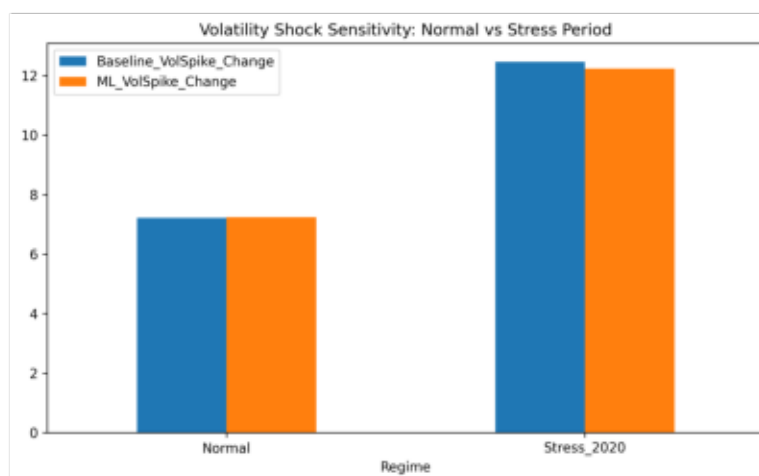


Figure 4 compares volatility-shock sensitivity in normal and stress regimes.

Figure 5. Combined Shock by Regime

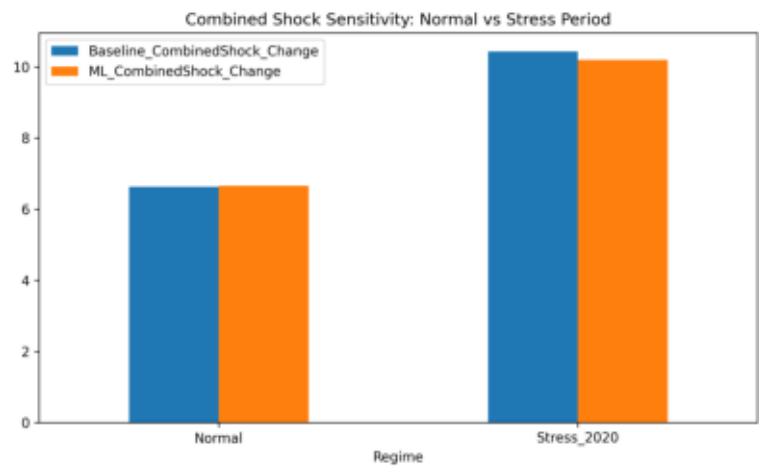


Figure 5 shows combined-shock sensitivity, where the volatility effect remains dominant despite partial rate offset.

5. Machine Learning Volatility Prediction

The selected ML route is indirect pricing: first predict future realized volatility, then feed the predicted sigma back into the BSM chooser pricing framework. This route preserves financial interpretability because the ML model changes only the volatility input rather than replacing the pricing formula with a black-box price predictor.

Candidate models included linear regression, random forest, and gradient boosting. After tuning, gradient boosting was selected because it delivered the strongest tuned validation and test balance among the compared volatility models. Its performance suggests that nonlinear interactions among recent volatility, VIX behavior, and rate features contain useful predictive signal.

The SHAP and feature-importance evidence indicates that recent volatility-related variables dominate the forecast. This is financially plausible: near-term realized volatility, rolling stock volatility, and market-risk variables tend to carry the strongest information for future realized volatility.

Table 6. Volatility Model Comparison

Source: outputs/tuned_vol_model_comparison.csv

The tuned comparison is preferred because it represents the selected final model search.

Model	Validation_MAE	Validation_RMSE	Validation_R2	Test_MAE	Test_RMSE	Test_R2
Best Random Forest	0.002500	0.004600	0.9490	0.003200	0.007800	0.8940
Best Gradient Boost	0.002400	0.004400	0.9529	0.003200	0.007200	0.9091

Figure 6. Gradient Boosting SHAP Bar Summary

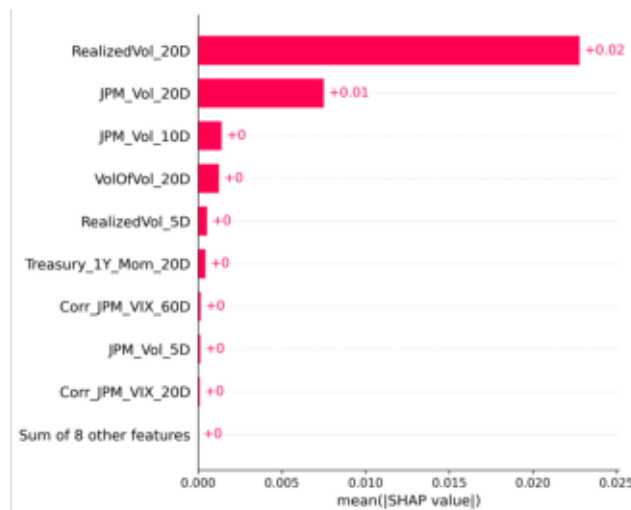


Figure 6 ranks the most important features in the selected gradient boosting volatility model.

Figure 7. Gradient Boosting SHAP Beeswarm

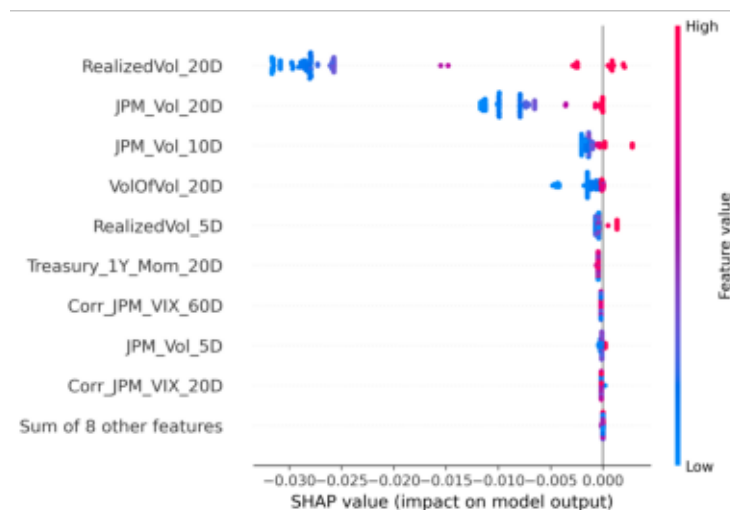


Figure 7 shows feature-level direction and dispersion of the SHAP effects across observations.

6. ML-Enhanced BSM Pricing Results

The final predicted volatility was converted into an annualized sigma and fed into the chooser pricing function. This created two comparable pricing series: baseline BSM chooser prices and ML-enhanced BSM chooser prices. Benchmark evaluation then compared the models against available option-market marks.

The key empirical conclusion is nuanced. Machine learning improves volatility prediction, but this does not translate into materially better benchmark pricing inside the current BSM structure. In the final benchmark summary, the ML-enhanced BSM route is close to the baseline but does not dominate it on the headline MAE/RMSE metrics.

The supplementary diagnostics help explain the result. Improvements are mixed across observations and regimes, tail behavior remains meaningful under shock scenarios, and sigma-gap diagnostics show that even a correlated sigma adjustment can produce limited price gains. The pricing error appears to be driven by more than historical or realized-volatility forecasting alone.

Regime-specific significance results should be interpreted cautiously because some refined subperiods contain too few benchmark observations for reliable formal inference.

Table 7. Benchmark Pricing Error Summary

Source: outputs/final_benchmark_error_summary.csv

Model	MAE	RMSE
Baseline BSM	4.5876	7.3348
GBR-Enhanced BSM	4.9346	8.2907

Table 8. Supplementary Significance Tests

Source: outputs/week7_significance_tests.csv

Group	N	MeanImprovement	MedianImprovement	ImprovementRatio	Paired_t_pvalue	Wilcoxon_pvalue
All	22	-0.1217	0.0451	0.5909	0.7401	0.3278
Crash_2020	2					
Normal	18	0.0549	0.0451	0.6111	0.1610	0.2475
Recovery_2020	2					

Table 9. Tail Statistics under Shock Scenarios

Source: outputs/week7_tail_statistics.csv

Scenario	Mean	Std	P95	P99	Max	Share_gt_20pct
Baseline_VolSpike	0.2538	0.1735	0.4950	0.4976	0.4989	0.5431
ML_VolSpike	0.2549	0.1731	0.4952	0.4977	0.4988	0.5384
Baseline_RateHike	-0.0220	0.0505	0.0670	0.1121	0.1501	0.0000
ML_RateHike	-0.0223	0.0492	0.0657	0.0997	0.1160	0.0000
Baseline_Combined	0.2318	0.1832	0.4883	0.4978	0.5060	0.5033
ML_Combined	0.2325	0.1828	0.4882	0.4970	0.5045	0.5057

Table 10. Regime Tail Diagnostics

Source: outputs/week7_regime_tail_statistics.csv

Regime	Baseline_VolSpike	Baseline_VolSpike	Baseline_VolSpike	ML_VolSpike_Mean	ML_VolSpike_P95	ML_VolSpike_Share_gt20
Crash_2020	0.3103	0.3673	1.0000	0.3224	0.3645	1.0000
Late2020	0.1746	0.3903	0.4353	0.1712	0.3886	0.4235
Normal	0.2596	0.4957	0.5343	0.2613	0.4957	0.5301
Other_2020	0.3490	0.3968	1.0000	0.3492	0.3964	1.0000
Recovery_2020	0.1943	0.3303	0.4811	0.1872	0.3297	0.4717

Table 11. Sigma-Gap Summary

Source: outputs/week7_sigma_gap_summary.csv

Statistic	SigmaGap
count	1681.0000
mean	-0.000300
std	0.0161
min	-0.1562
25%	-0.005200
50%	-0.000100
75%	0.004700
max	0.2399

Table 12. Sigma Correlation Matrix

Source: outputs/week7_sigma_correlation.csv

Variable	BaselineSigma_Annual	FinalPredictedSigma_Annual
BaselineSigma_Annual	1.0000	0.9950
FinalPredictedSigma_Annual	0.9950	1.0000

Table 13. Sigma Gap and Price Gap Correlation

Source: outputs/week7_sigma_price_gap_correlation.csv

This result shows that sigma differences do map into price differences mechanically, but the sigma adjustments are usually too small to generate large economic repricing gains.

Variable	AbsSigmaGap	AbsPriceGap
AbsSigmaGap	1.0000	0.9438
AbsPriceGap	0.9438	1.0000

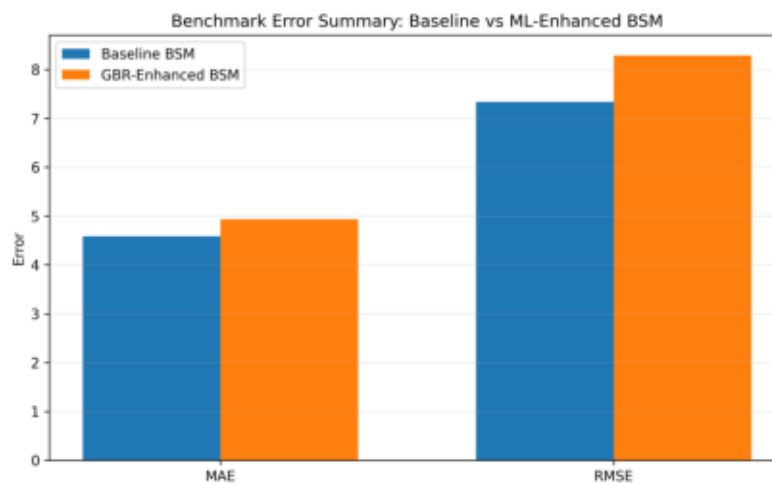
Figure 8. Benchmark Error Summary

Figure 8 compares baseline and ML-enhanced BSM benchmark pricing errors.

Figure 9. Improvement Distribution

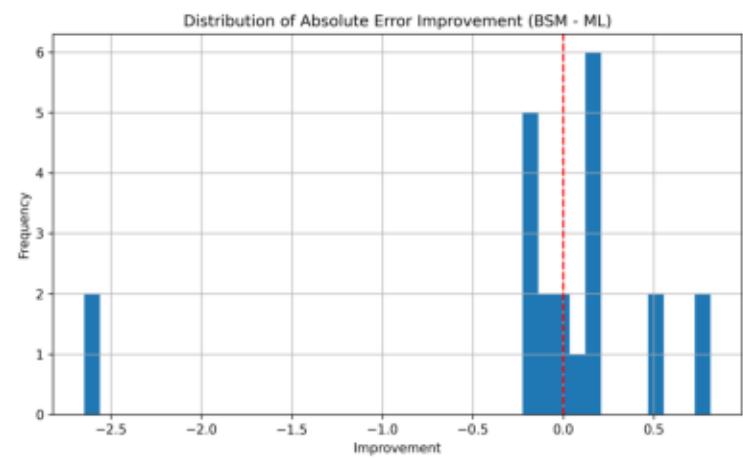


Figure 9 shows that ML-enhanced pricing gains are mixed rather than uniformly positive.

Figure 10. Improvement by Regime

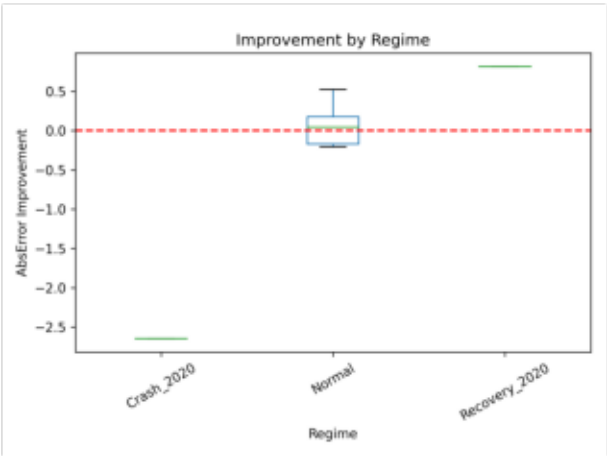


Figure 10 summarizes average improvement across regime groups.

Figure 11. Sigma Gap versus Price Gap

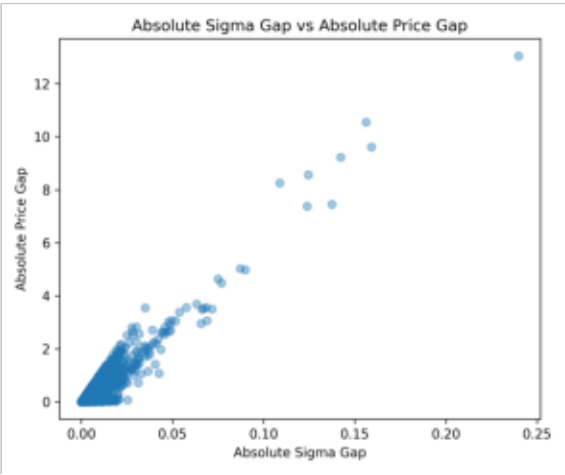


Figure 11 links volatility-input differences to resulting chooser-price differences.

Figure 12. Tail Exceedance

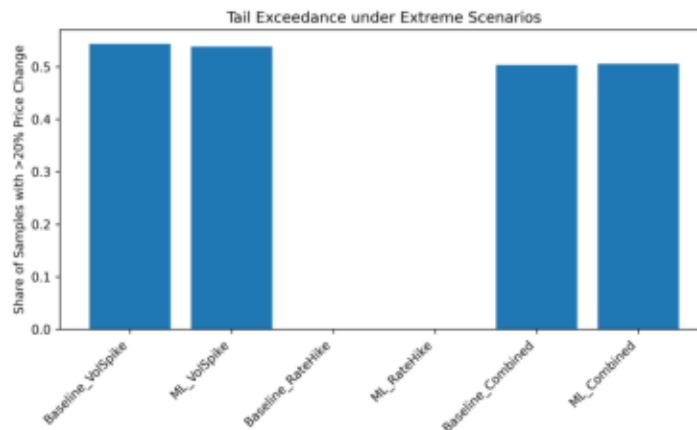


Figure 12 reports the share of shock samples with more than 20 percent price change.

7. Tool Prototype and Data Integration

The project includes a Streamlit-based pricing dashboard that turns the modeling workflow into an interactive prototype. The dashboard is designed around latest loaded data, contract and market inputs, dual pricing under baseline and ML-enhanced BSM assumptions, an error-margin or adjustment summary, quick scenario checks, and visual dashboard output.

The local data-loading module supports the tool by reading the latest processed project data. It should be interpreted as an automated project-data integration layer rather than a true live market-data feed. This distinction matters: the prototype is operational for reproducible local analysis, but production deployment would require authenticated live data sources, validation checks, and more robust update scheduling.

Figure 13. Streamlit Pricing Dashboard Prototype



Figure 13 shows the local dashboard prototype used to expose the pricing workflow interactively.

8. Discussion

The main finding is that a better volatility forecast is not automatically a better option price. Realized volatility prediction and option-market pricing are related, but they are not the same target. Market option prices embed implied volatility, risk premia, liquidity, supply-demand pressure, jump expectations, skew, term structure, and contract-specific microstructure effects.

Because the ML enhancement changes sigma only, it cannot correct all structural limitations of the BSM framework. The BSM assumptions of lognormal diffusion, constant parameters, frictionless markets, and simplified dividend treatment are restrictive for real option benchmarks. A sigma-only update is therefore useful as a controlled experiment, but insufficient as a full market-pricing engine.

Future improvements should add implied-volatility surface features, option-volume and spread information, sentiment or macro-risk variables, liquidity and market-microstructure features, and richer model structures that can represent skew, term structure, and jump risk. A second future route is hybrid modeling: preserve arbitrage-aware pricing constraints while learning residual corrections from option-market data. The central lesson is therefore not that machine learning is ineffective, but that a sigma-only enhancement is too narrow to overcome the broader structural mismatch between historical-volatility forecasting and market option pricing.

9. Conclusion

The project successfully implemented a baseline chooser option pricing workflow, integrated real-world data and engineered features, trained and selected an ML volatility model, fed ML-predicted sigma into the BSM pricing framework, stress-tested the pricing response, and delivered a usable local prototype tool.

The empirical results are clear: chooser prices are highly sensitive to volatility shocks, stress regimes amplify sensitivity, and gradient boosting provides a strong volatility-prediction route. However, ML-enhanced sigma does not materially improve benchmark option pricing under the current BSM structure. The strongest conclusion is therefore not that ML failed, but that volatility-only enhancement is too narrow for market option pricing. The prototype and data module provide a practical foundation for the next stage: implied-volatility-aware, residual-corrected, and production-grade pricing workflows.